THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 7: **Work, Energy and Simple Machines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): No work is done when a person pushes against a rigid wall. Reason (R): The displacement of the wall is zero. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A body is thrown upward with KE of 200 J. At half the maximum height the KE is (ignoring air resistance): 1

(a) 50 J **(b)** 200 J **(c)** 0 **(d)** 100 J

3. A block of mass 5 kg is pulled up a smooth incline of length 10 m to a height of 5 m. The work done against gravity is (g = 10): 1

(a) 25 J **(b)** 250 J **(c)** 500 J **(d)** 50 J

4. A person of mass 60 kg runs up a flight of 20 stairs, each 20 cm high, in 5 s. Power is (g = 10): 1

(a) 960 W **(b)** 480 W **(c)** 240 W **(d)** 120 W

5. Assertion (A): The kinetic energy of a body becomes four times when its speed is doubled. Reason (R): KE is proportional to the square of the speed. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A body falls from rest from a height h. At what height is its speed half of the final speed? 1

(a) h/2 **(b)** h/4 **(c)** h/8 **(d)** 3h/4

7. A bob of a simple pendulum is displaced and released from 0.8 m height above the lowest point. The speed at the lowest point is: 1

(a) 16 m/s **(b)** 2 m/s **(c)** 8 m/s **(d)** 4 m/s

8. A 100 W bulb and a 2 kW heater are used for 3 hours and 1 hour respectively. The ratio of energy consumed is: 1

(a) 3 : 2 **(b)** 20 : 3 **(c)** 1 : 20 **(d)** 3 : 20

9. **Assertion (A):** No work is done by a person who holds a heavy suitcase while standing still. 1
Reason (R): Work is done only when a force produces a displacement of the body in the direction of the force.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The joule is the SI unit of energy. 1
Reason (R): Power is the rate of doing work.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A body of mass 2 kg is acted on by a force that changes its speed from 3 m/s to 5 m/s. Find the work done by the force. 2

12. An engine develops 2 x 10^5 W power and moves a train at 20 m/s. Find the force of the engine. 2

13. Why does the PE of a body not depend on the path taken to reach a height? 2

14. A 1 kg ball is thrown vertically up with a KE of 100 J and rises to only 8 m. Find the energy lost to air resistance (g = 10). 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A block of mass 5 kg slides down a rough incline of height 4 m and reaches the bottom at 6 m/s. Find the work done against friction (g = 10). 3

16. A spring stores 4 J of energy when stretched by 2 cm. If the energy is proportional to the square of the extension, how much energy does it store when stretched by 4 cm? 3

17. An object of mass 2 kg is lifted vertically from rest by a force of 30 N through 3 m. Find the work done by the force, by gravity and the final speed (g = 10). 3

OR

A 60 kg man climbs a 5 m high ladder in 10 s carrying a 20 kg load. Find the work done and the average power (g = 10). 3

18. A motor lifts a 200 kg elevator through 30 m at a constant speed in 20 s with 60 per cent efficiency. Find the input power (g = 10). 3

OR

A body of mass 3 kg is thrown upward at 20 m/s. When its KE is half the initial KE, find the height and the speed (g = 10). 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Ravi, a student of mass 50 kg, climbs the stairs of his school building through a vertical height of 10 m in 20 s. Take $g = 10$ m/s^2. 5
(i) Find the gain in his potential energy.
(ii) How much work does he do against gravity?
(iii) Find his power.
(iv) If he climbs the same stairs in 40 s, what is his power, and does the work done change?

20. A car of mass 1200 kg accelerates from rest at 1.5 m/s squared for 10 s against a constant resistive force of 600 N. Find 5
(a) the final speed
(b) the KE
(c) the work done by the engine
(d) the average power
(e) the final power.

OR

A man pulls a cart up a slope with a rope; the cart of mass 80 kg rises 3 m over a slope length of 12 m. The tension is 250 N along the slope. Find the work by the tension, the PE gained, the work against friction and the efficiency (g = 10). 5

END OF QUESTION PAPER Total: 40 Marks